

Homework 2: Inventory Management System ER Diagram

Assigned on: Jan 23, 2026

Due on: Jan 28, 2026

Submit your ER diagram to Canvas

1. Find Entities
2. Find Relationships
3. Draw (handwritten) ER schema diagram and submit.
4. **Optional:** You can **also** submit an ER diagram by using a modeling tool like ERWin.

You have been tasked with designing a database for an inventory management system for a retail store. The store needs to track its products, suppliers, store locations, and sales transactions. Use the following information and design a Conceptual Data Model using ER Diagram for the given inventory management system.

1. **Products:** Each product in the store is uniquely identified by a product ID and has a name. Products may have multiple suppliers.
2. **Suppliers:** Each supplier is identified by a supplier ID and has a name and contact information (address, phone number, email). A supplier can supply many products.
3. **Store Locations:** The store operates in multiple locations, each identified by a unique location ID. Each location has an address. Product can be available at many or no location. And a location can have many products
4. **Inventory:** The store maintains inventory records for each product available in each store location. Each product in each location has a stock quantity and a price.
5. **Sales Transactions:** Whenever a customer purchases a product from a store location, a sales transaction is recorded. Each transaction includes the product sold, the quantity sold, the sale price, the date of sale, and the customer's details.